

Files

main

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akshitkdesign

Added test.ipynb file

d7902f7 · 2 weeks ago

History

Preview

Code

Blame

156 lines (156 loc) · 3.7 KB

Raw

Python Programming Test

Test Date: 18 February 2026

Q1. Stable Top-K Selection

```
In [ ]: def top_k(items, k):
        if k <= 0:
            return []

        indexed_items = list(enumerate(items))
        indexed_items.sort(key=lambda t: (-t[1][1], t[0]))
        return [item for _, item in indexed_items[:k]]

items = [("A", 90), ("B", 95), ("C", 95), ("D", 85)]
print(top_k(items, 2))
```

Q2. Category Word Counter

```
In [ ]: def count_by_category(pairs):
        result = {}
        for cat, word in pairs:
            if cat not in result:
                result[cat] = {}
            result[cat][word] = result[cat].get(word, 0) + 1
        return result

pairs = [("sports", "win"), ("sports", "win"), ("tech", "ai")]
print(count_by_category(pairs))
```

Q3. Missing Number (0 to n)

```
In [ ]: def missing_number(nums):
        n = len(nums)
        total = n * (n + 1) // 2
        return total - sum(nums)

nums = [3, 0, 1]
print(missing_number(nums))
```

Q4. Default Argument Bug

```
In [ ]: def add_unique(x, seen=None):
        if seen is None:
            seen = []
        if x not in seen:
            seen.append(x)
        return seen

print(add_unique(1))
print(add_unique(2))
```

Q7. Longest Unique Substring

```
In [ ]: def longest_unique_substring(s):
        last_pos = {}
        left = 0
        best = 0

        for right, ch in enumerate(s):
            if ch in last_pos and last_pos[ch] >= left:
                left = last_pos[ch] + 1
            last_pos[ch] = right
            best = max(best, right - left + 1)

        return best

print(longest_unique_substring("abcabcbb"))
```